US financial markets, oil market shocks, and the COVID-19 pandemic

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I. Introduction

The relationship between oil prices and financial markets is a focus of empirical macroeconomics. There is a high level of variation of oil prices over time and the resulting effect receives significant attention from policymakers (Klepacz, 2021). Shocks in the oil industry are typically divided into demand and supply shocks. Demand shocks are caused by changes in the business cycle (aggregate demand) or shifts in market expectations (precautionary demand). Supply shocks result from disruption or change to the amount or availability of oil. The latter often results from geopolitical conflicts and tends to be more covered by news outlets, despite demand shocks being more impactful to U.S. growth trajectories (Kilian, 2010). Both supply and demand shocks impact financial markets.

In this paper, I examine oil market shocks and S&P 500 returns using monthly data from 1990 to 2025. I will employ a structural vector autoregression (SVAR) following the approach of Kilian (2009). In this approach, shocks are divided into supply, aggregate demand, and oil-specific demand shocks (precautionary demand shocks). Kilian's (2009) results indicate that aggregate and precautionary demand shocks are the most significant and persistent for macroeconomic outcomes. Kilian and Park (2009) extend this framework to US stock markets and show that financial market reactions differ across this division of shocks. They find that supply shocks have a negligible impact on returns and aggregate and precautionary demand shocks are the primary driver of returns.

Sakaki (2019) examines heterogeneity of the US stock market and argues that insignificant effects are driven by differing sectorial reactions and thus, supply shocks do drive returns. Additionally, high oil price volatility hurts returns for all sectors. Filis et al., (2011) show that the relationship between financial markets and oil prices is not static and shifts based on market conditions. Moreover, during periods of economic instability, oil markets do not hedge against equity market losses.

My findings for the 1990 - 2025 period indicate that aggregate demand shocks have the most significant effect on financial markets. Supply shocks and precautionary demand shocks have little effect on equity markets. Hence, overall economic conditions matter most.

Further, oil price volatility causes temporary market reactions but not persistent effects.

While the SVAR framework provides a robust identification for shocks, Degiannakis et al. (2018) argue that linear SVAR models may not capture any time-varying dynamics. To address this limitation, I adopt a sub-sample estimation approach to isolate the COVID-19 period which marked a change in economic regime. I find that this period altered the way financial markets reach to oil market shocks. Specifically, markets have stronger, more erratic reactions to shocks showing an increased sensitivity and volatility. Moreover, these results suggest that the relationship between commodity markets and financial markets continues to evolve.

II. Data

Data is monthly for the period of January 1990 to December 2025. This period includes several shocks to oil markets including the Asian financial crisis (1997), the great recession (2007 - 2009), the US shale revolution (2007 - 2017), the COVID-19 pandemic (2020 - 2023), and the Russian invasion of Ukraine (2022). In total, there are 432 observations.

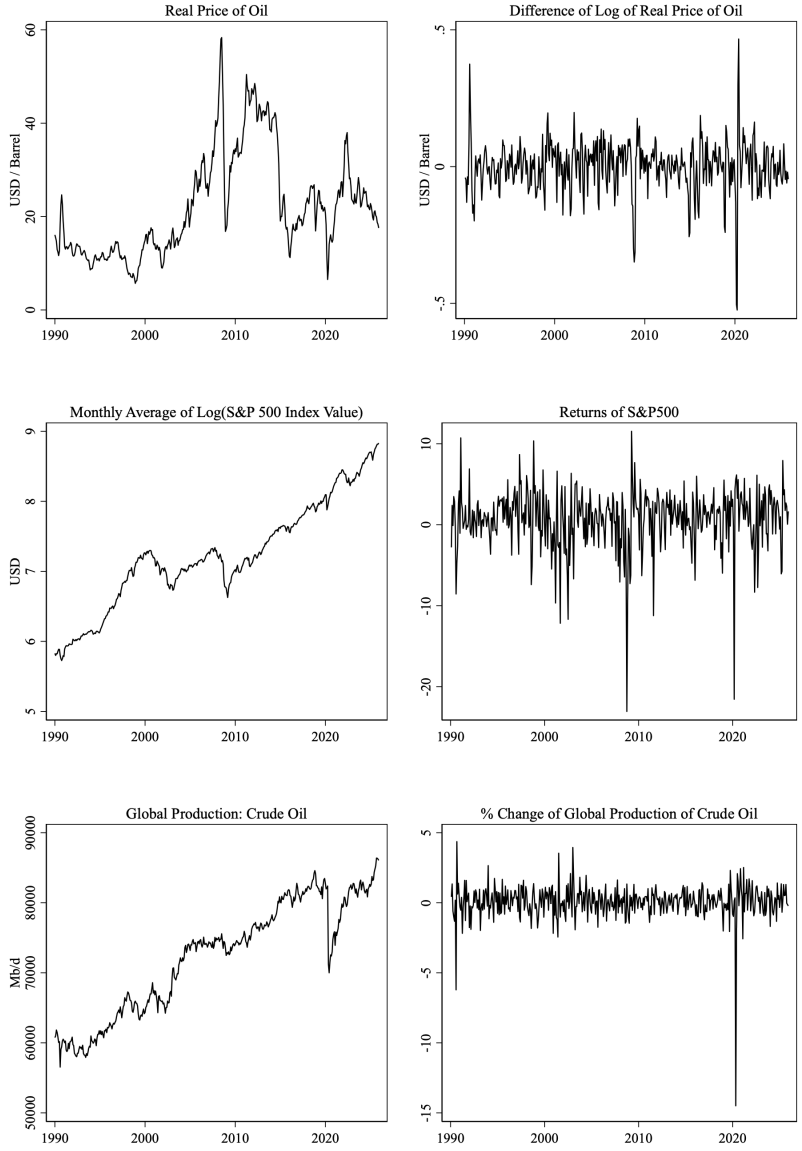
I source daily index values for the S&P 500 (ŜPX) from S&P Capital IQ. From the daily prices, I calculate the monthly arithmetic mean of the log of the daily prices. Finally, I take the difference multiplied by 100 to obtain the monthly percentage returns of the index.

Oil market shocks are captured in three time series. First, global oil production is given by total world crude oil production in millions of barrels per day (Mb/d) sourced from the U.S. Energy Information Administration (EIA). To make the series stationary, I take the percent change of global oil production. Second, aggregate demand is observed using Kilian's (2009) index of global real economic activity (REA). This index captures business-cycle fluctuations in global demand for industrial commodities by using changes in bulk dry cargo shipping, expressed as percentage deviations from the overall trend.

Lastly, the real price of oil (RPO) is measured from the acquisition cost of imported U.S. crude oil for refiners sourced from the EIA and adjusted for inflation using the consumer price index (CPI) sourced from the U.S. Bureau of Labor Statistics. Due to the shutdown of the US government in October 2025, CPI data is unavailable for this month. Therefore, I have linearly interpolated the CPI value using September and November 2025. To make this series stationary, I take the difference of the log of RPO.

Figures 1 and 2 provide the time series for each of the four variables as discussed above.

Figure 1: Time series for RPO, S&P 500, and Global Production of Oil



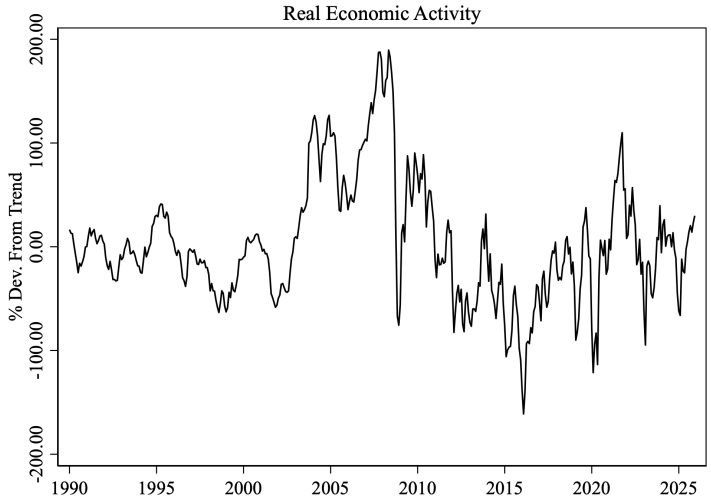


Figure 2: REA

Additionally, I consider the volatility of the price of oil using the daily price of West Texas Intermediate (WTI) sourced from EIA. I calculate the monthly volatility as the standard deviation of the logged daily returns, scaled by the square root of 21 which is the average number of trading days within a month. Figure 2 plots the time series for monthly volatility of WTI.

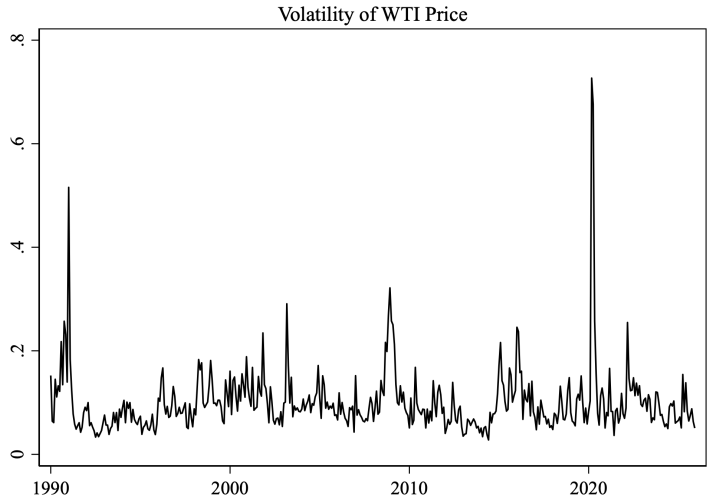


Figure 3: WTI Monthly Volatility

Table 1 provides descriptive statistics and Table 2 provides unit root tests for the five variables. A modified Dicky-Fuller test for stationarity is included in Appendix C.

Table 1: Descriptive Statistics

	Mean	Median	Max	Min	Std. Dev.	Skewness	Kurtosis	N
Real Economic Activity (REA)	2.816	-5.499	189.528	-161.177	58.860	0.739	3.878	432
% Change Global Production	0.081	0.170	4.364	-14.491	1.194	-4.358	55.144	431
Diff. Log(RPO)	0.000	0.009	0.466	-0.524	0.092	-0.748	9.474	431
S&P 500 Returns (%)	0.697	1.095	11.546	-23.049	3.628	-1.601	10.654	431
WTI Volatility	0.100	0.087	0.727	0.028	0.064	5.038	42.721	432
Observations	432							

Entire sample period 1990 - 2025

Table 2: Unit Root Tests

	ADF(t)	ADF(p)	PP(t)	PP(p)
Panel A				
REA	-4.094	0.001	-3.692	0.004
Monthly Average of Log(S&P500)	-0.240	0.934	-0.119	0.947
RPO	-3.138	0.024	-2.486	0.119
Global Oil Production	-0.876	0.796	-0.822	0.813
Panel B				
% Change Global Production	-16.312	0.000	-21.553	0.000
Diff. Log(RPO)	-13.957	0.000	-12.496	0.000
S&P 500 Returns	-13.987	0.000	-17.052	0.000
WTI Volatility	-9.609	0.000	-10.735	0.000

1. Panel A gives unit root tests for pre- adjusted variables and shows all are non-stationary with the exception of REA. RPO is almost stationary but fails to reject the null hypothesis under the PP test.
2. Panel B provides results after adjusting variables and shows all are stationary.
3. ADF: Augmented Dicky Fuller test
4. PP: Phillips-Perron test

III. Empirical Approach

I employ a Structural Vector Autoregression Model (SVAR) model to examine the dynamic impact of oil market shocks on financial markets. Specifically, following the framework established by Kilian (2009), I decompose supply and demand shocks to assess their distinct effects on the S&P 500.

Model Specification

Let Y_t denote a vector of endogenous variables measured at monthly frequency:

$$Y_t = \begin{pmatrix} \Delta prod_t \\ REA_t \\ \Delta rpo_t \\ R_t \end{pmatrix}$$

where $\Delta prod_t$ represents the % change in global production of crude oil, REA_t is the index of real economic activity, Δrpo_t gives the difference of the log of the real price of oil, and R_t represents the return of the S&P 500 all at time t. I estimate the following structural VAR for the period between January 1990 and December 2025:

$$A_0 Y_t = \alpha + \sum_{i=1}^{p} A_i Y_{t-i} + \epsilon_t \quad (1)$$

where ϵ_t represents the vector of structural shocks, which are assumed to be white noise and orthogonal. The optimal lag length, p is chosen based on the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) to ensure there is no serial correlation in the residuals.

Identification Strategy

The relationship between the reduced-form residuals (e_t) and the structural shocks (ϵ_t) is given as follows:

$$e_t = A_0^{-1} \epsilon_t.$$

To identify the structural shocks, I impose a recursive exclusion restriction using a Cholesky decomposition of the variance-covariance

matrix. Specifically, I assume that the matrix A_0^1 is lower triangular, as in the below:

$$e_t \equiv \begin{pmatrix} e_{1t}^{\Delta prod_t} \\ e_{2t}^{REA_t} \\ e_{3t}^{\Delta rpo_t} \\ e_{4t}^{R_t} \end{pmatrix} = \begin{pmatrix} b_{11} & 0 & 0 & 0 \\ b_{21} & b_{22} & 0 & 0 \\ b_{31} & b_{32} & b_{33} & 0 \\ b_{41} & b_{42} & b_{43} & b44 \end{pmatrix} \begin{pmatrix} \epsilon_{1t}^{\Delta prod_t} \\ \epsilon_{2t}^{REA_t} \\ \epsilon_{3t}^{\Delta rpo_t} \\ \epsilon_{4t}^{R_t} \end{pmatrix}$$

The above ordering reflects the assumption that oil production does not respond to changes in aggregate demand in a given month due to high adjustment costs associated with production. In other words, oil supply is exogenous within a given month. Globabl economic activity (REA) responds immediately to changes in the supply but responds to changes in oil prices with a one month lag. This aligns with the fact supply immediately affects shipping costs and factory inputs while sticky behavior of firms and consumers delays the reaction to changes in oil prices. The oil prices themselves respond contemporaneously to changes in production of oil and aggregate demand. Lastly, financial markets respond instaneously to all innovations across oil markets.

IV. Results

Model Selection

Table 3 presents the results of the length test to select p, the optimal lag length for (1). A lag length of five minimizes the AIC while a length of one minimizes the BIC. Given these results and that a one month lag is not sufficient to capture persistence in commodity markets, I select a VAR(5) model. Additional selection tests are included in Appendix A.

The results from the Granger causality test in Table 4 show a dynamic, bi-directional relationship between oil markets and the S&P 500. Specifically, the tests indicate that financial markets also serve as an indicator for commodity markets. This result does not contradict the recursive structural identification. Instead, it suggests that while supply and demand shocks impact equity prices immediately, financial markets are also forward looking and thus, provide predictive information on how commodity markets will adjust in future months.

Table 3: Lag Length Test Results

Lag	AIC	BIC
0	17.468	17.506
1	14.830	**15.022**
2	14.735	15.081
3	14.737	15.237
4	14.737	15.390
5	**14.687**	15.494
6	14.710	15.670
7	14.713	15.827
8	14.741	16.009
9	14.720	16.141
10	14.746	16.321

[1.] Minimizing values of each criterion are in bold.

Table 4: Granger Causality Wald Tests

	Equation	Excluded	chi2
REA			
d_lglobalpro	0.972	5.000	0.965
d_lRPO	26.861	5.000	0.000
d_lSP500	13.790	5.000	0.017
ALL	48.538	15.000	0.000
d_lglobalpro			
REA	12.862	5.000	0.025
d_lRPO	40.783	5.000	0.000
d_lSP500	25.334	5.000	0.000
ALL	95.221	15.000	0.000
d_lRPO			
REA	10.488	5.000	0.063
d_lglobalpro	5.344	5.000	0.375
d_lSP500	16.542	5.000	0.005
ALL	31.145	15.000	0.008
d_lSP500			
REA	45.344	5.000	0.000
d_lglobalpro	3.863	5.000	0.569
d_lRPO	10.336	5.000	0.066
ALL	51.118	15.000	0.000

Impulse Response Functions (IRFs)

Figure 4 shows the reactions of financial markets to different types of oil market shocks. Positive shocks to global aggregate demand trigger a statistically significant increase in the S&P 500, which peaks in the second month following the shock. However, by months five to six, there is a statistically significant reversal, where the response dips below the baseline. This could suggest a stage of market correction following the initial expansionary phase. Additionally, this result could be attributed to a lagged impact of increased input costs or anticipation of monetary tightening in response to increased economic activity. This counteracting effect is persistent, with the response remaining below the zero threshold for the entire 24 month horizon at a statistically significant level.

In contrast, the S&P 500 shows no statistically significant response to global production shocks. While there is some evidence of a positive response to supply innovation, it is not significant and returns to near zero by month eight. Hence, supply disruptions or surpluses do not signal a change in returns. This could be driven by the existence of global strategic reserves or anticipation of offsetting market decisions by swing producers such as OPEC.

Lastly, the S&P 500 response to oil price innovation is largely statistically insignificant. There is evidence of an oscillatory response but this is not significant. This result suggests that there may be some nominal volatility as a result of oil price changes, but this impact is not robust enough to distinguish from market noise.

Overall, these results support the conclusion that shocks to aggregate demand are the most reliable predictor of equity market returns.

Figure 4: IRFs for Entire Sample Period

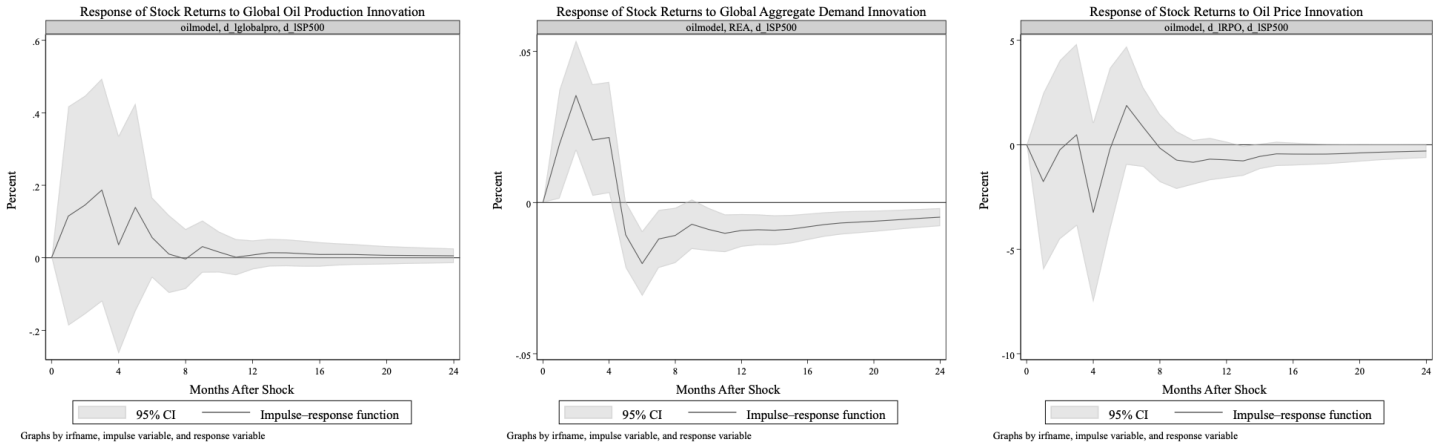


Table 5: Forecast Error Variance Decomposition of S&P 500 Returns

Step	MSE	Global Oil Production (%)	Aggregate Demand (%)	Oil Price (%)	Stock Returns (%)
0	0.000	0.00	0.00	0.00	0.00
2	3.401	0.29	1.18	2.85	95.68
4	3.497	0.85	5.17	2.75	91.23
6	3.548	1.07	6.33	3.03	89.57
8	3.575	1.06	7.50	3.15	88.29
10	3.583	1.07	7.87	3.16	87.91
12	3.593	1.07	8.28	3.20	87.45
14	3.602	1.07	8.66	3.24	87.03
16	3.610	1.08	9.01	3.25	86.66
18	3.616	1.08	9.27	3.27	86.39
20	3.620	1.08	9.46	3.28	86.18
22	3.624	1.08	9.62	3.29	86.01
24	3.627	1.08	9.74	3.29	85.88

[1.] Values represent the percentage of forecast error variance explained by shocks to global oil production, global economic activity (REA), oil prices, and stock returns.
[2.] MSE denotes the mean squared forecast error.

Volatility of Oil Prices

To consider volatility of oil prices, I substitute WTI monthly volatility for Δrpo_t following Sakaki (2019). For details of the model, see Appendix B.

As can be seen in Figure 5, a shock to oil price volatility produces an immediate spike in financial markets but this spike is ultimately transitory, as the effect becomes statistically indistinguishable from zero for the remainder of the time horizon. This result in combination with Figure 4 suggests that markets may experience some fluctuations in response to oil market news from geopolitical events or that influences precautionary demand, but these effects are not persistent.

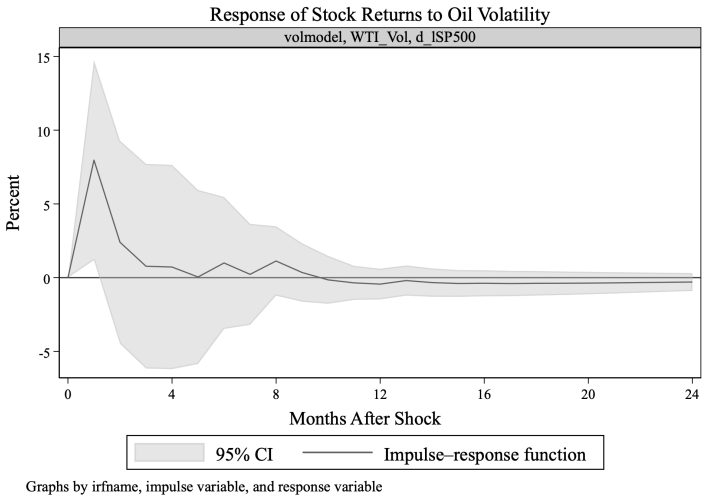


Figure 5: IRF for Volatility

Table 6: Forecast Error Variance Decomposition of S&P 500 Returns (Volatility Model)

Step	MSE	Global Oil Production (%)	Aggregate Demand (%)	Oil Price Volatility (%)	Stock Returns (%)
0	0.000	0.00	0.00	0.00	0.00
2	3.429	0.24	0.94	9.43	89.39
4	3.507	0.78	3.82	9.36	86.04
6	3.554	1.39	4.91	9.17	84.52
8	3.577	1.38	6.03	9.08	83.50
10	3.588	1.38	6.55	9.07	83.00
12	3.599	1.38	7.06	9.01	82.54
14	3.607	1.37	7.46	8.97	82.19
16	3.614	1.37	7.78	8.94	81.91
18	3.619	1.37	8.02	8.92	81.69
20	3.623	1.37	8.20	8.90	81.53
22	3.626	1.37	8.33	8.89	81.41
24	3.628	1.37	8.43	8.88	81.31

[1.] Values represent the percentage of forecast error variance explained by shocks to global oil production, global economic activity (REA), oil price volatility (WTI volatility), and stock returns.
[2.] MSE denotes the mean squared forecast error.

COVID-19 Period

Figures 6-9 below compare financial market responses to shocks before and after the COVID-19 pandemic, which was an unprecedented negative aggregate demand shock. As this event could alter the relationship between oil markets and financial markets, I divide the sample

into pre-COVID and post-COVID periods. Hence, I can assess if transmission changed following the pandemic.

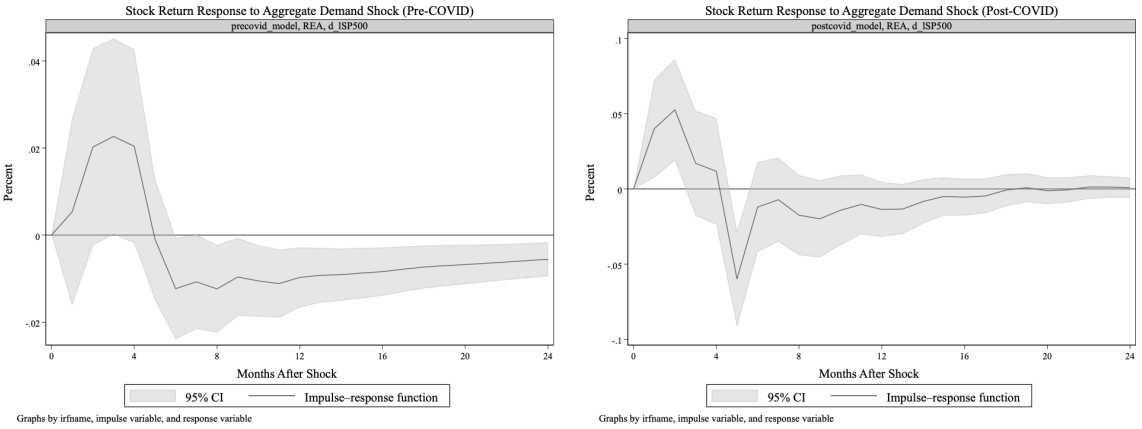


Figure 6: Aggregate Demand Shocks in Pre and Post-COVID Periods

Both periods follow a similar pattern of an initial positive response to aggregate demand innovation followed by a negative response. However, the post pandemic impulse response are more significant, and show greater volatility. Additionally, the self-correcting, negative response is much sooner than in the pre-pandemic period. Additionally, in the pre-period, aggregate demand shocks appear to be more persistent while in the post-period, they appear transient. These result could be attributable to increased uncertainty which may have amplified transmission of shocks to financial markets.

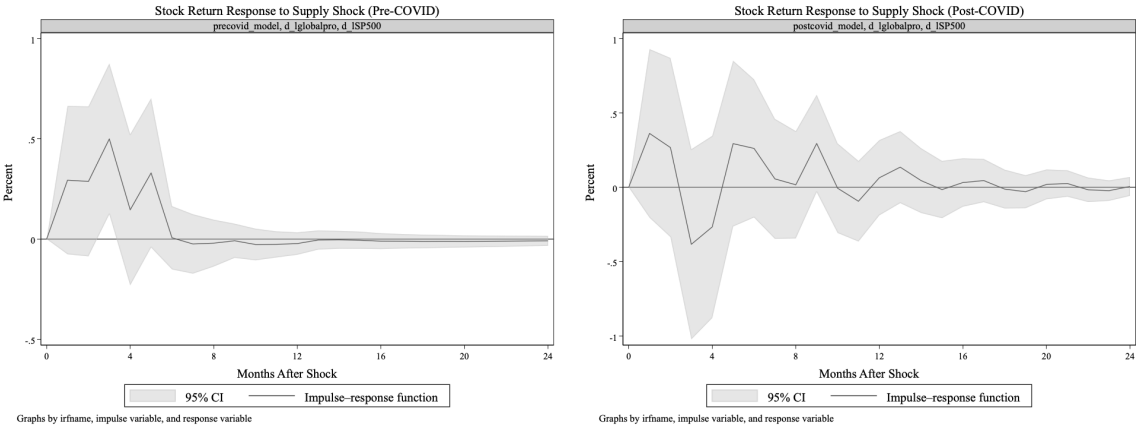


Figure 7: Supply Shocks in Pre and Post-COVID Periods

For supply shocks, the pattern between the pre and post-pandemic eras is more contrasting. In the pre-period, supply shocks had a significant, positive effect on financial markets whereas during the post-period, this reaction is more volatile. Although the results are noisy, this indicates that financial markets may be more sensitive to supply chain disruption since the pandemic.

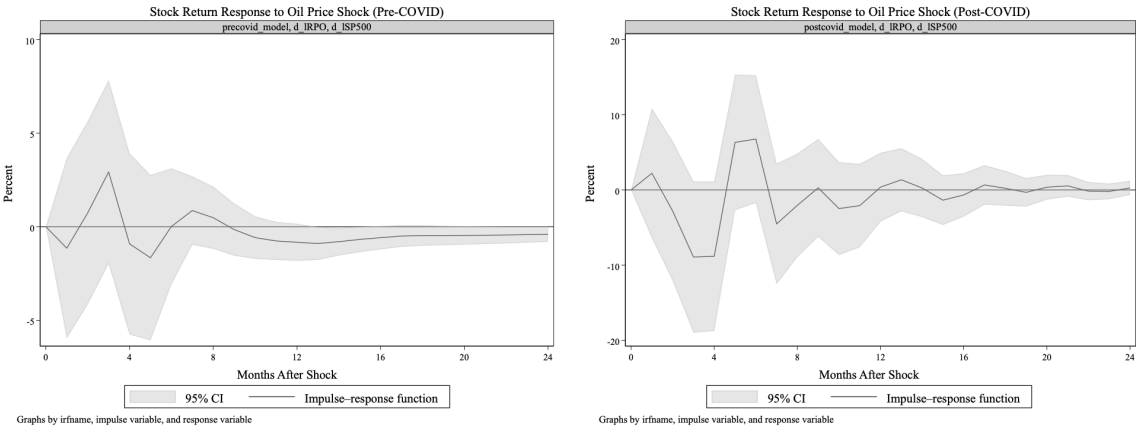


Figure 8: Oil Price Shocks in Pre and Post-COVID Periods

For oil price shocks prior to the pandemic, financial market responses were generally insignificant. In the post-period, there is a more distinct ”V” shape in the response, although still insignificant. Overall, this suggests that average financial market responses to oil price shocks are more volatile and unpredictable post-pandemic.

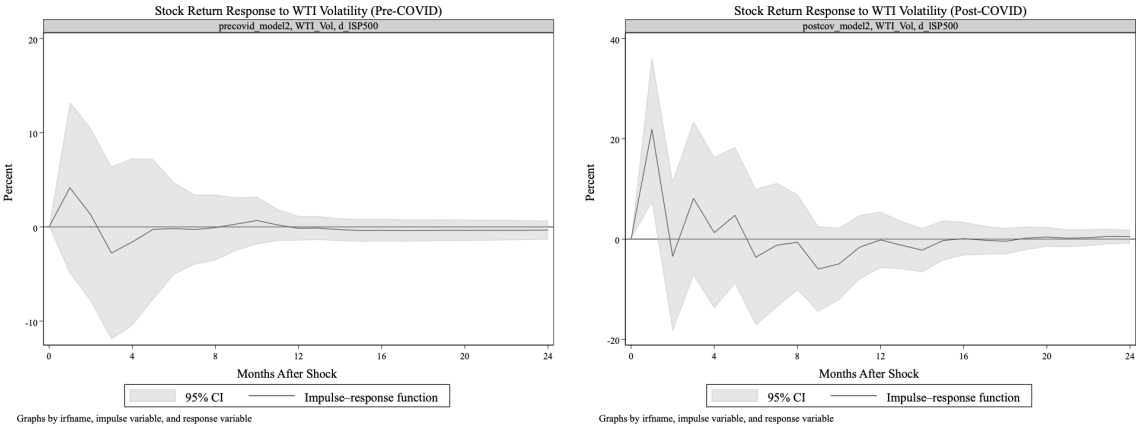


Figure 9: Oil Price Volatility Shocks in Pre and Post-COVID Periods

A similar pattern is apparent for financial market response to volatility. Pre-pandemic, the response was limited and insignificant. Post-pandemic, financial markets became more sensitive to oil price volatility. The initial positive response becomes larger and significant in the post period.

Overall, this sub-sample analysis indicates a shift in financial market responses following the COVID-19 pandemic. Across all types of shocks, impulse responses from the S&P 500 are more significant and oscillatory.

V. Conclusion

Overall, the results indicate that aggregate demand shocks play the most consistent role in equity market returns. Shocks to global oil production and oil prices have largely insignificant effects. This indicates that economic conditions are the main driver through which oil markets influence financial markets. Oil price volatility generates a response in equity markets, but it is short-lived. Hence, price volatility may only contribute to temporary fluctuations as opposed to long-term changes.

Using a subsample analysis, I find that the relationship between oil and equity markets has evolved since the COVID-19 pandemic. Financial market responses have become more volatile and unpredictable, which may indicate increased sensitivity to economic uncertainty. These results align with the view that the relationship between financial markets and commodity markets evolves with economic conditions.

These findings support existing literature showing that demand shocks are the most important link between oil and equity markets. Additionally, they highlight the necessity of considering structural changes in economic conditions when assessing this relationship. Although this framework provides a useful identification strategy for shocks, future research could explore a model that allows for time-varying relationships or non-linear dynamics. Additionally, the analysis could be extended for sector-level or international variation.

VI. References

Federal Reserve Bank of Dallas. (n.d.). *Index of global real economic activity (IGREA)*. Retrieved 31 March 2026, from https://www.dallasfed.org/research/igrea

Filis, G., Degiannakis, S., & Floros, C. (2011). Dynamic correlation between stock market and oil prices: The case of oil-importing and oil-exporting countries. *International review of financial analysis, 20*(3), 152-164.

Kilian, L. (2009). Not all oil price shocks are alike: Disentangling demand and supply shocks in the crude oil market. *American economic review, 99*(3), 1053-1069.

Kilian, L., & Park, C. (2009). The impact of oil price shocks on the US stock market. *International economic review, 50*(4), 1267-1287.

Kilian, L. (2010). *Oil price volatility: Origins and effects* (No. ERSD-2010-02). WTO Staff Working Paper.

Klepacz, M. (2021). Price setting and volatility: Evidence from oil price volatility shocks. *International Finance Discussion Paper*, (1316).

Lüthkepohl, H. (2006). New Introduction to Multiple Time Series Analyses.

Sakaki, H. (2019). Oil price shocks and the equity market: Evidence for the S&P 500 sectoral indices. *Research in International Business and Finance, 49*, 137-155.

S&P Global Market Intelligence. (n.d.). *S&P 500 Index Value (Daily, 1990 - 2025)*. Retrieved 17 March 2026, from Capital IQ database.

U.S. Bureau of Labor Statistics. (n.d.). *Consumer price index for all urban consumers: All items in U.S. city average [CPIAUCSL]*. Federal Reserve Bank of St. Louis. Retrieved 17 March 2026, from https://fred.stlouisfed.org/series/CPIAUCSL

U.S. Energy Information Administration. (n.d.). *U.S. Crude Oil Imported Acquisition Cost by Refiners, monthly*. Retrieved 31 March 2026, from https://www.eia.gov/dnav/pet/hist/LeafHandler.ashx?n=PET&s=R1300____3&f=M

U.S. Energy Information Administration. (n.d.). *Crude oil prices: West Texas Intermediate (WTI) - Cushing, Oklahoma [DCOILWTICO]*. Federal Reserve Bank of St. Louis. Retrieved 17 March 2026, from https://fred.stlouisfed.org/series/DCOILWTICO

U.S. Energy Information Administration. (n.d.). *World - crude oil including lease condesate production, monthly*. Retrieved 01 April 2026, from https://www.eia.gov/international/data/world/petroleum-and-other-liquids/monthly-pet

roleum-and-other-liquids-production?pd=5&p=00000000000000000000000000000002&u=0&f=M&v=line&a=-&i=none&vo=value&vb=173&t=C&g=none&l=249249--&l=249--249&s=94694400000&e=1764547200000&ev=true

VII. Appendices

Appendix A: VAR Model Selection Details

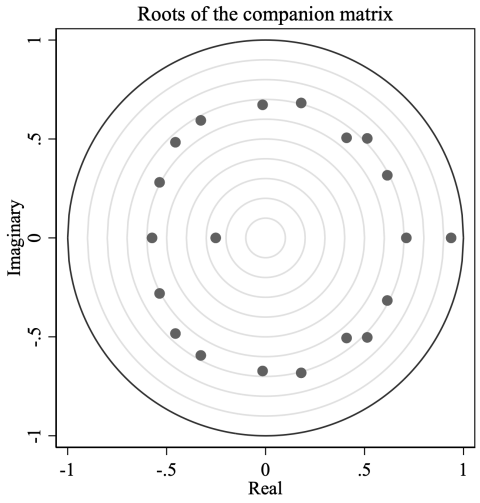
The following tables and figure provide further details on the selection and justification of the VAR(5) specification.

Table 7: Residual Serial Correlation Lagrange Multiplier Test

	Lag	chi2	df	Prob > chi2
1	1.000	17.302	16.000	0.366
2	2.000	17.540	16.000	0.351
3	3.000	24.854	16.000	0.072
4	4.000	35.679	16.000	0.003
5	5.000	14.985	16.000	0.526

The above Lagrange Multiplier test generally supports the specification of the VAR(5) model as the majority of the lags have p-values above the 5% significance level. This result indicates an absence of serial correlation in the residuals. Although the null is rejected at Lag 4, since most lag orders do not reject the null hypothesis, the model appears broadly well specified.

Figure 10: Eigenvalue Stability Test



The above graph shows that ll eigenvalues lie within the unit circle which means the VAR(5) model satisfies the stability condition and is suitable for impulse response analysis.

Table 8: Residual Normality Tests (Jarque-Bera)

	norm_test		
	Equation	chi2	df
REA	127.009	2.000	0.000
d_lglobalpro	6579.974	2.000	0.000
d_lRPO	109.338	2.000	0.000
d_lSP500	256.520	2.000	0.000
ALL	7072.842	8.000	0.000

Although the normality tests reject the null hypothesis that the residuals are normally distributed, this is a standard result for high frequency data such as oil and equity markets. Lütkepohl (2006) notes that the least squares estimators used for impulse response functions do not rely on the assumption of normally distributed residuals if the sample size is sufficiently large and the model is stable. Specifically, the least squares estimators remain asymptomatically consistent and normally distributed.

Appendix B: VAR Model Selection Details for Inclusion of WTI Volatility

The following tables provide details on the selection of the VAR model used S&P 500 impulse response functions for oil price volatility.

Table 9: Lag Length Test Results

Lag	AIC	BIC
0	16.737	16.776
1	13.849	**14.041**
2	13.748	14.094
3	13.737	14.236
4	13.722	14.375
5	**13.671**	14.477
6	13.692	14.652
7	13.681	14.795
8	13.708	14.976
9	13.693	15.115
10	13.727	15.302

[1.] Minimizing values of each criterion are in bold.

Table 10: Granger Causality Wald Tests

	Equation	Excluded	chi2
REA			
d_lglobalpro	0.894	5.000	0.971
WTI_Vol	5.966	5.000	0.310
d_lSP500	12.925	5.000	0.024
ALL	26.642	15.000	0.032
d_lglobalpro			
REA	11.811	5.000	0.037
WTI_Vol	94.832	5.000	0.000
d_lSP500	21.725	5.000	0.001
ALL	155.574	15.000	0.000
WTI_Vol			
REA	29.003	5.000	0.000
d_lglobalpro	9.884	5.000	0.079
d_lSP500	7.488	5.000	0.187
ALL	52.252	15.000	0.000
d_lSP500			
REA	37.730	5.000	0.000
d_lglobalpro	3.988	5.000	0.551
WTI_Vol	7.606	5.000	0.179
ALL	48.132	15.000	0.000

Table 11: Residual Serial Correlation Lagrange Multiplier Test

	Lag	chi2	df	Prob ¿ chi2
1	1.000	13.479	16.000	0.637
2	2.000	30.841	16.000	0.014
3	3.000	21.774	16.000	0.151
4	4.000	30.923	16.000	0.014
5	5.000	11.143	16.000	0.801

Figure 11: Eigenvalue Stability Test

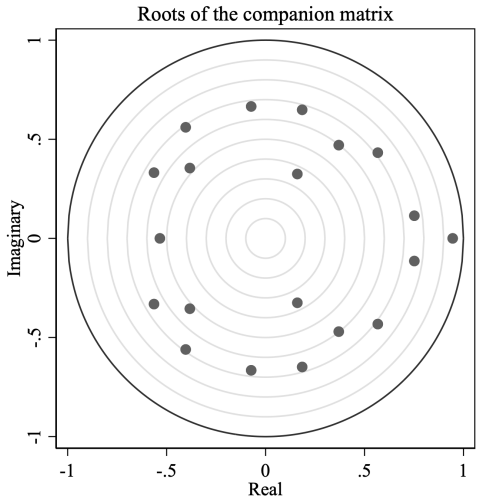


Table 12: Residual Normality Tests (Jarque-Bera)

	Equation	chi2	df
REA	126.083	2.000	0.000
d_lglobalpro	849.281	2.000	0.000
WTI_Vol	46366.054	2.000	0.000
d_lSP500	358.557	2.000	0.000
ALL	47699.975	8.000	0.000

Appendix C: Additional Test on Variable Stationarity

Table 13 tests for if the variables are stationary using the modified Dicky-Fuller Test. The results generally align with the augmented Dicky-Fuller (ADF) and Phillips-Perron (PP) tests presented in Table 2. However, S&P500 returns and REA are only stationary at the 10% confidence-level. Additionally, using this test unadjusted global oil production is stationary. However, given that the literature generally uses oil production growth and the ADF and PP tests find this variable to be non-stationary, my model uses production growth.

Table 13: Modified Dicky-Fuller Test (DF-GLS)

Variable	**DF-GLS Tau**	**Critical Value (1%)**	**Critical Value (5%)**	**Critical Value (10%)**
RPO	-2.069	-3.480	-2.828	-2.545
d_lRPO	-4.406	-3.480	-2.818	-2.536
SP500	-1.637	-3.480	-2.866	-2.579
d_lSP500	-2.724	-3.480	-2.818	-2.536
REA	-2.558	-3.480	-2.823	-2.541
GLOBAL_OIL_PROD	-3.098	-3.480	-2.828	-2.545
d_lglobalpro	-4.341	-3.480	-2.818	-2.536
WTI_Vol	-5.725	-3.480	-2.873	-2.586